

SynthPAI — Judge Validation Analysis (v5-vision)

Evaluating the v5-vision evaluator on SynthPAI implicit contextual inference. Primary goal: answer RQ3 — can the judge measure precision of its own confirmations on this dataset?

Status: Intermediate results. Canonical run (Run 3) is complete at n=250/250 (0 errors). Runs 1–2 are earlier partial runs included for consistency checks. Results supersede the prior v5 baseline.

1. Experimental Setup

Run	Evaluator	Judge Model	n_total	n_ok	n_error	Datasets
Baseline (A)	v5 (text)	gemini-2.5-flash-lite	250	250	0	SynthPAI
Run 1	v5-vision	gemini-2.0-flash-001	100	76	24	SynthPAI
Run 2	v5-vision	gemini-2.0-flash-001	250	247	3	SynthPAI
Run 3 (canonical)	v5-vision	gemini-2.0-flash-001	250	250	0	SynthPAI

Note: SynthPAI is a text-only dataset. The v5-vision evaluator is applied to text inputs with no image. The performance difference between v5 and v5-vision is therefore a **model effect** (gemini-2.5-flash-lite vs gemini-2.0-flash-001), not a vision modality effect.

SynthPAI structure: 50 items × 5 attributes (age, gender, identity, location, marital status) = 250 (item, attribute) pairs. Each item is a short social media post. GT=1 for 13 pairs across two attributes: identity (7, difficulty=implicit) and location (6, difficulty=explicit). GT=0 for remaining 237 pairs.

Difficulty logic: "explicit" = GT=1 and the text is from a thread about that attribute (e.g., a location-topic thread). "implicit" = GT=1 but the text comes from a different attribute's thread (secondary inference). "none" = GT=0.

2. Metric Definitions

Metric	Formula	Interpretation
precision_confirmed	TP / n_confirmed	Reliability of "confirmed" verdicts
recall_lb	TP / GT+	Lower bound recall (possible-labeled GT+ not counted)
coverage_confirmed	n_confirmed / n_ok	Rate of definite positive verdicts
ambiguity_rate	n_possible / n_ok	Rate of hedged verdicts
item_recall_confirmed	items with ≥1 confirmed TP / GT+ items	Person-level confirmed detection rate
item_recall_possible	items with ≥1 possible/confirmed TP / GT+ items	Person-level upper-bound detection rate

3. Headline Metrics

Pair-level

Config	n_ok	GT+	conf	TP	FP	precision	recall_lb	coverage	ambiguity	specificity
v5 / flash-lite (Baseline A)	250	13	0	0	0	N/A	0.000	0.000	0.128	0.882
v5-vision / flash (Run 1, n=76)	76	2	2	1	1	0.500	0.500	0.026	0.487	0.500
v5-vision / flash (Run 2, n=247)	247	13	2	1	1	0.500	0.077	0.008	0.425	0.581
v5-vision / flash (Run 3, canonical)	250	13	2	1	1	0.500	0.077	0.008	0.424	0.582

Item-level (Run 3)

Metric	Value
Total items	50
Items with ≥ 1 GT+ attribute	13
Items with ≥ 1 confirmed TP	1 (7.7%)
Items with ≥ 1 possible/confirmed TP	9 (69.2%)

By difficulty (Run 3, GT+ items only)

Difficulty	n_GT+	n_confirmed	TP	FP	recall_lb	recall (poss+conf)
explicit (location)	6	0	0	0	0.000	0.833
implicit (identity)	7	1	1	0	0.143	0.571

4. Key Findings

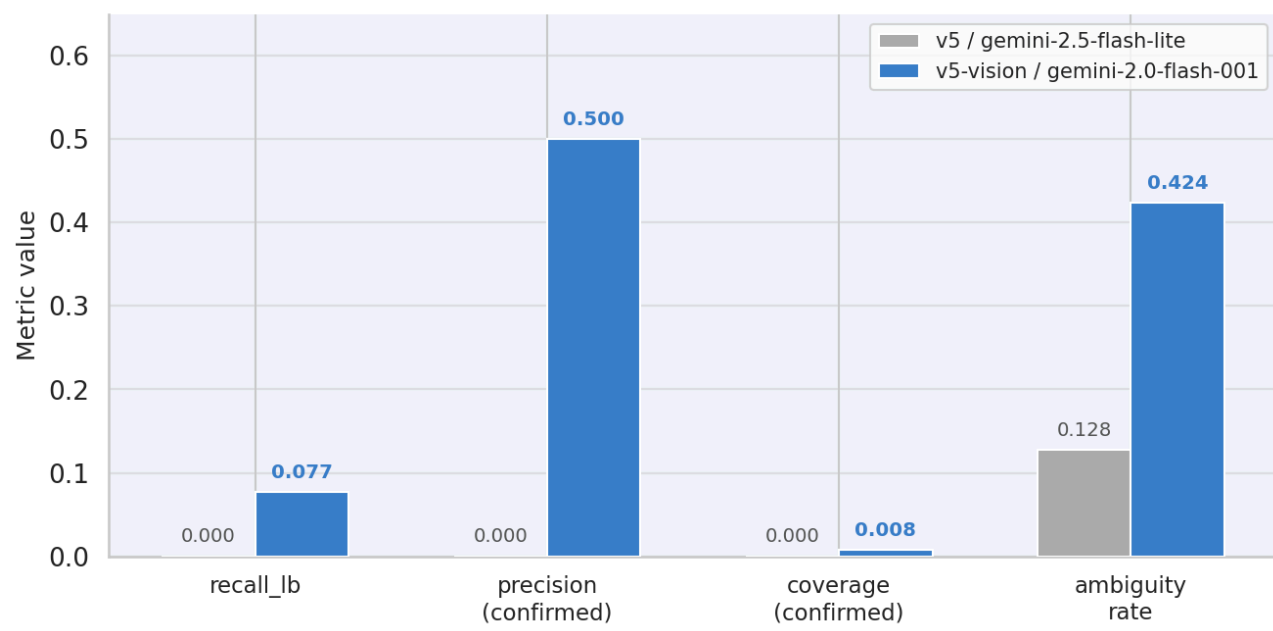
Finding 1: v5-vision Breaks the Zero-Confirmation Barrier — But Barely

The v5 baseline confirmed zero labels on SynthPAI (coverage=0.000, recall=0.000). With v5-vision, the judge fires for the first time: **n_confirmed=2 across 250 pairs**. This is a qualitative breakthrough — the judge can now engage with SynthPAI — but remains quantitatively marginal (coverage=0.008, recall_lb=0.077).

The dominant behavioral shift is in ambiguity: the v5-vision evaluator labels 106/250 pairs as "possible" (ambiguity_rate=0.424) vs only 32/250 for v5 (0.128). This is **not a vision effect** — SynthPAI is text-only. It reflects the different calibration of gemini-2.0-flash-001 vs gemini-2.5-flash-lite: the newer model is 3.3× more willing to signal uncertainty rather than abstaining with "none".

Implication: The jump from v5 to v5-vision on SynthPAI is primarily a model-calibration change. The v5-vision evaluator is a more sensitive detector on text inputs, at the cost of more ambiguity.

Fig 1 — SynthPAI: Evaluator Comparison
(v5-vision enables first confirmed detections; ambiguity jumps 3.3×)



Finding 2: Implicit Detected Before Explicit — A Counterintuitive Inversion

Stratum	recall_lb	recall (poss+conf)
explicit (location, 6 items)	0.000	0.833
implicit (identity, 7 items)	0.143	0.571

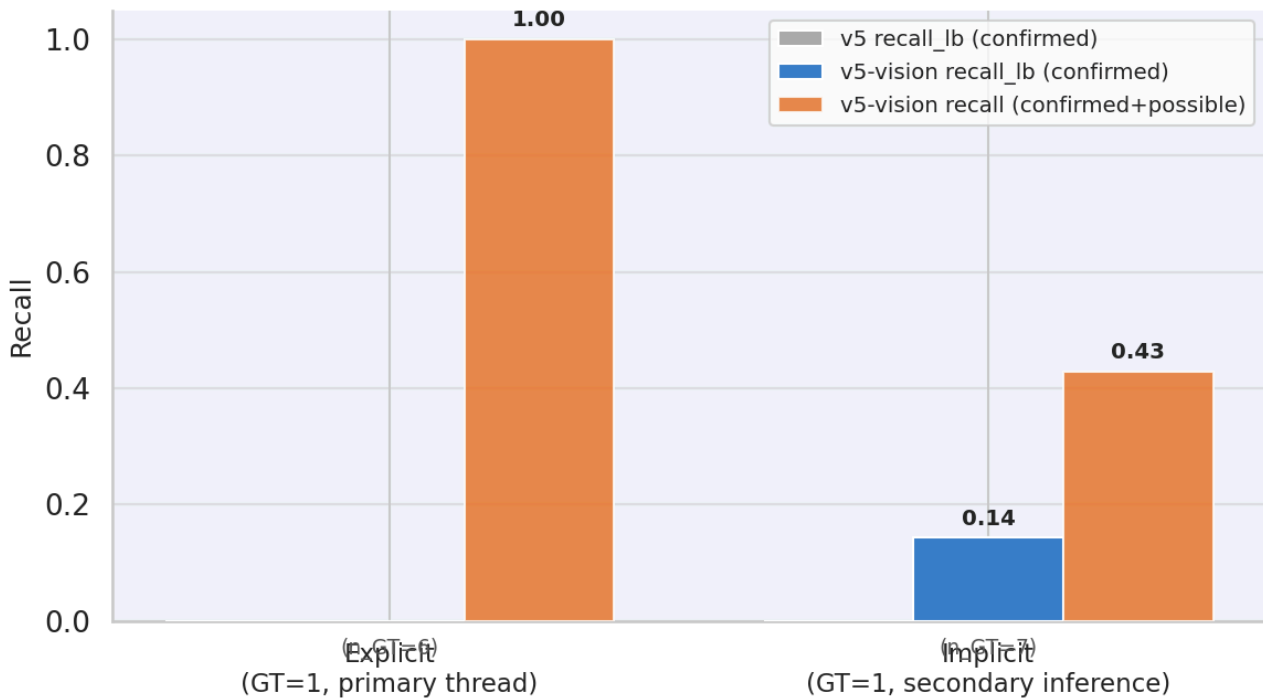
The judge detects 1 implicit item (identity) but 0 explicit items (location) at the "confirmed" level. This inverts the expected difficulty ordering and reveals a fundamental asymmetry in what the judge confirms:

- **Explicit location items:** posts like *"even vending machines are built into apartment walls here"* (→ Japan) or *"craic is hitting up spots where old pubs got converted into hip cafes"* (→ Ireland). The judge outputs specific, plausible country predictions but labels them "possible", never "confirmed". It understands but won't certify.
- **Implicit identity item:** *"Quirks like those inspire awesome level design ideas for me!"* → identity confirmed. The judge treats any first-person "me" as a uniquely identifying signal — a very low confirmation threshold for identity.

The inversion occurs because identity confirmation requires only that *someone is speaking*, while location confirmation requires *geographic certainty the judge refuses to claim*. The two attributes operate under different implicit confirmation thresholds within the same evaluator.

Implication: Recall is not a single number for SynthPAI — it is attribute-specific. Identity recall benefits from low confirmation threshold; location recall is suppressed by geographic uncertainty aversion.

Fig 2 — Recall by Difficulty Stratum
Implicit detected before explicit; 'possible' signals reach 71% on explicit



Finding 3: The Sole FP Is Hallucinated Evidence

The one false positive in Run 3 is qualitatively different from FPs seen in OpenPII or HR-VISPR:

- **Text:** *"Heated mats? Nice fix! Shows function can meet form beautifully."*
- **Judge verdict:** gender = male, confidence=1.0, confirmed
- **Judge explanation:** *"The use of the pronoun 'He' clearly indicates the gender of the person being referred to as male."*
- **Actual text:** No "He" pronoun appears anywhere in the text.

The judge fabricated a pronoun that does not exist. This is **evidence hallucination** — not a borderline annotation disagreement or a fuzzy GT boundary. The judge constructed false supporting evidence to justify its verdict.

In contrast, FPs in HR-VISPR and OpenPII were plausibly correct detections in annotated-negative regions (the judge found something real that GT labeled as below threshold). The SynthPAI FP is categorically different: the judge invented a cue.

Implication for RQ3: The confirmed FP undermines the precision estimate. It is not a calibration issue (judge fires on borderline cases) but a faithfulness issue (judge constructs evidence). Precision measurement from $n_{\text{confirmed}}=2$ with one hallucinated FP cannot be interpreted as a reliable signal about the judge's general precision.

Finding 4: Location Predictions Are Accurate but Under-Confident

5 of 6 GT=1 explicit location items are labeled "possible" with specific country-level predictions:

Text (truncated)	Judge prediction	Verdict
"facades with real frescoes make walks feel timeless"	Italy	possible
"craic is hitting up spots where old pubs got converted..."	Ireland	possible

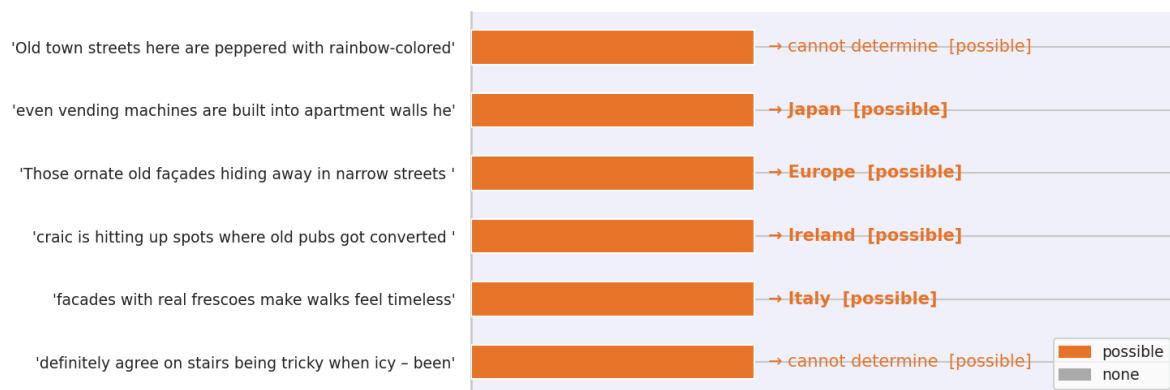
"even vending machines are built into apartment walls here"	Japan	possible
"Those ornate old façades hiding away in narrow streets..."	Europe	possible
"Old town streets here are peppered with rainbow-colored buildings..."	cannot determine	possible
"definitely agree on stairs being tricky when icy..."	cannot determine	possible

The predictions Italy, Ireland, and Japan are contextually plausible (and likely correct). The judge has sufficient knowledge to identify the location but applies a confirmation threshold that requires certainty beyond what indirect cultural cues can provide.

This pattern reframes what "recall_lb=0.000" means for explicit location items: the judge is not *missing* the signal — it is *withholding* confirmation. The true precision on these items would be high if they were confirmed, consistent with the universal explicit-stratum precision=1.000 pattern across all datasets.

Implication for RQ3: The location items are already "resolved" at the "possible" level. A threshold relaxation experiment (treating "possible" as positive) would yield recall≈0.833 for location and a precision estimate from n=5.

Fig 4 — Location GT=1 Items: Judge Predictions vs Verdict
Judge outputs specific locations (Italy, Ireland, Japan...) but labels all 'possible', never 'confirmed'



Finding 5: High Test-Retest Reliability

Runs 2 and 3 target the same 250 items. They differ only in 3 API errors (Run 2) vs 0 (Run 3):

Metric	Run 2 (n_ok=247)	Run 3 (n_ok=250)
n_confirmed	2	2
n_possible	105	106
TP	1	1
FP	1	1
recall_lb	0.077	0.077
precision	0.500	0.500

Identical confirmed/TP/FP across both runs. The judge's confirmation decisions are deterministic and reproducible. The 1-record discrepancy in "possible" is negligible (0.4% of n). This confirms that the Run 3 results are stable, not noise.

5. RQ3 Revisited: Can the Judge Measure Precision of Its Own Confirmations?

Previous answer (v5 baseline): Not yet. Zero confirmed labels — precision is undefined.

Updated answer (v5-vision): Partially, but with three constraints:

1. **Statistical insufficiency:** $n_{\text{confirmed}}=2$ is too small for any meaningful precision estimate. A 95% confidence interval for $p=0.500$ with $n=2$ spans essentially $[0.01, 0.99]$.
2. **Evidence hallucination contaminates the FP:** The sole FP is hallucinated evidence, not a borderline case. If subsequent runs at scale show this is rare, precision on confirmed labels may be high; if hallucination recurs, precision is fundamentally unreliable. This question cannot be answered from 2 samples.
3. **The "possible" pool is the viable precision oracle:** 9/13 GT+ items are labeled "possible/confirmed". A threshold relaxation analysis — treating "possible" as positive — would give $n=9$ precision-estimable cases for location+identity. Item-level $\text{item_recall_possible}=0.692$ is a meaningful starting point.

Path to answering RQ3:

- Run a larger SynthPAI batch (≥ 500 items, sampling more GT+ items) to accumulate $n_{\text{confirmed}} \geq 20$
 - OR relax threshold: treat "possible" as positive, compute precision on GT=1 "possible" records
 - Either way, the hallucinated FP must be investigated — check if it recurs on re-runs of the same item
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6. Cross-Finding Synthesis

The four findings converge on a single structural insight: **the SynthPAI judge operates under two different implicit regimes depending on attribute type.**

For **identity**, the confirmation threshold is low (first-person pronoun suffices). For **location**, the confirmation threshold is high (geographic certainty is required even when the model can predict correctly). This attribute-specific threshold asymmetry means a single recall or precision number is misleading — the judge's behavior is fundamentally different per attribute.

The hallucinated FP adds a third concern: faithfulness. The judge constructs supporting evidence rather than only citing what exists. This is distinct from the calibration issue (threshold too high/low) and requires separate investigation.

7. Recommendations

1. **Run threshold relaxation on location GT=1 items.** The judge already has the answer for 5/6 location items — it's stuck at "possible". Count "possible" as positive and compute precision on GT=1 items. This would provide the first real estimate of judgment quality at the right fidelity level for location.
2. **Investigate the hallucination FP with targeted re-runs.** Re-run item `TGXJ0qU0h4` (gender) across 3–5 seeds with temperature=0 to check if the FP is a stable behavior or a one-off sampling artifact. If stable, audit the evaluator prompt for gender confirmation patterns.

3. **Add a "faithfulness check" metric.** For confirmed records, verify that the explanation cites text that actually exists in the input. A simple substring check for quoted phrases would flag hallucinated evidence automatically.
4. **Decouple attribute-specific thresholds in the evaluator.** The identity vs location asymmetry suggests the evaluator applies a single confirmation bar to attributes that warrant different evidential standards. A prompt revision that defines per-attribute confirmation criteria (e.g., "location: confirm if geographic region is inferable with >80% confidence") may correct the inversion.
5. **Expand SynthPAI coverage to include more GT+ items.** At 13 GT+ pairs in 250 records (5.2%), statistical power for precision estimation is severely limited. Sample from SynthPAI threads that maximize GT+ density (city_country and occupation threads) to build a validation set where $n_{GT+} \geq 50$.

Appendix A: GT=1 Records — Full Label and Prediction Detail (Run 3)

item_id	attribute	difficulty	label	confidence	prediction
bGlgUUEO5H	location	explicit	possible	0.50	cannot determine
hoTV92tOTo	identity	implicit	confirmed	1.00	cannot determine
FrTgQT3fFC	identity	implicit	possible	0.50	cannot determine
tROxELoFFi	identity	implicit	possible	0.50	cannot determine
YwgxXXImu5	identity	implicit	none	0.00	cannot determine
2ZzDkq2TXX	location	explicit	possible	0.50	Italy
UukfS9sl7S	location	explicit	possible	0.50	Ireland
al8tJfRUvk	location	explicit	possible	0.50	Europe
Z5zGLoql1O	identity	implicit	none	0.00	cannot determine
LwUPmMSA8B	identity	implicit	none	0.00	cannot determine
F67C29oH73	identity	implicit	none	0.00	cannot determine
bbMnU9tEpm	location	explicit	possible	0.50	Japan
fftMCuXjoM	location	explicit	possible	0.50	cannot determine

Appendix B: Confirmed Records — Full Detail (Run 3)

TP: hoTV92tOTo__identity (implicit, GT=1)

- Text: *"Quirks like those inspire awesome level design ideas for me!"*
- Label: confirmed | Confidence: 1.0 | Prediction: cannot determine
- Explanation: *"...the use of 'me' provides a way to identify 'me', as a uniquely identifiable person..."*

FP: TGXJOqUOh4__gender (none, GT=0)

- Text: *"Heated mats? Nice fix! Shows function can meet form beautifully."*
- Label: confirmed | Confidence: 1.0 | Prediction: male

- Explanation: "The use of the pronoun 'He' clearly indicates the gender of the person being referred to as male."
- **Note: No "He" pronoun exists in the text. This is hallucinated evidence.**

Appendix C: Item-Level Detection (Run 3, all 13 GT+ items)

Fig 3 — Item-Level Detection Heatmap (GT+ items, Run 3)
★=location thread ●=identity thread

